



Maini Lotfi Abdelkader

Systems & Full Stack Developer

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Profile

A developer with experience in building web and mobile applications using React, Next.js, React Native, and Flutter. In addition to the modern web stack, I am a self taught systems programmer proficient in C and Rust. While I work across the full stack, my professional preference lies in Rust for its performance and memory safety. I am committed to writing clean, maintainable code and building modular tools from the user interface down to the system level.

Languages

Arabic – Native / Mother Tongue

Full command of reading, writing, and speaking; used in personal and professional communication.

French – Fluent, professional working proficiency

Able to communicate confidently in professional and technical contexts, including reading documentation and collaborating with French speaking colleagues.

English – Very good, strong written and spoken skills

Proficient in reading technical materials, writing code documentation, and communicating in professional or learning environments.

German – Currently learning, basic conversational level

Developing foundational reading, writing, and speaking skills to reach professional proficiency.

Education

MATHEMATICS BACCALAUREATE

Belkbire Abdelkader High School

- Completed secondary education with a focus on mathematics, developing strong analytical and problem-solving skills.

ATTENDED: TWO YEARS OF CIVIL ENGINEERING PROGRAM

Morsli Abdekah University

- Completed two years of civil engineering studies before transitioning to the tech field. Acquired foundational skills in technical reasoning, project planning, and logical problem solving, supporting a strong technical mindset for web development.

ADVANCED TECHNICIAN DIPLOMA IN WEB AND MOBILE DEVELOPMENT

National Institute of Specialized Vocational Training

- Specialized in full stack web and mobile development with hands on experience in HTML, CSS, JavaScript, React, Node.js, and PHP. Completed projects emphasizing responsive design, clean code, and user-centered applications.

Skills

Systems & Languages:

Low-Level : Rust, C, Java

Web/Core : TypeScript, JavaScript (ES6+), SQL (CLI-Proficient) , PostgreSQL

Environments : Linux (Debian/Arch), Windows (WSL2) , Podman

Frontend & Mobile:

Frameworks : React, Next.js, React Native, Flutter

Visuals : Three.js, GSAP, Responsive Design

UI/UX : HTML5, CSS3, ShadCN UI, Tailwind CSS

Backend & Infrastructure:

Runtimes/Engines : Node.js, Express.js, PHP

Databases : MySQL, MongoDB, Supabase, Drizzle ORM

Services : REST APIs, Clerk (Auth), JWT, MinIO (S3), Zod

Development Workflow:

Editors : Neovim, Vim, VS Code

Version Control : Git, GitHub

Utilities : Bash/Zsh, npm, Postman, Chrome DevTools

Professional Experience

Freelance Web & Mobile Developer

Self employed / Algeria

- Developed responsive websites and mobile applications for clients, using HTML, CSS, JavaScript, React, Node.js, and PHP.
- Created and maintained online stores and business websites, ensuring user-friendly design, functionality, and performance.
- Collaborated with clients to understand requirements, deliver projects on time, and implement iterative improvements.

Freelance Graphic Designer & 3D Modeling Artist

Self employed / Algeria

- Designed logos, banners, and visual content for web and print, meeting client branding requirements.
- Created 3D models and renders using Blender for various projects, including product visualization and creative presentations.

Projects

Linux System Monitor (C Systems)

Low Level Kernel Telemetry Tool

Github [↗](#)

(C, Linux Kernel, Procfs, ANSI/VT100)

- Engineered a high efficiency system monitor in C that interfaces directly with the Linux /proc virtual filesystem to extract real time hardware telemetry without external library overhead.
- Implemented a differential CPU analysis engine to calculate utilization metrics by processing kernel clock ticks and managing state deltas across high precision timed intervals.
- Optimized memory footprint and I/O performance by utilizing low level stream parsing and ANSI escape sequences to deliver a flicker free dashboard with near zero RAM allocation.

Rust TUI System Monitor

High Performance Terminal Telemetry Dashboard

Github [↗](#)

(Rust, Ratatui, Sysinfo, Crossterm)

- Engineered a stateful Terminal User Interface (TUI) using an immediate mode rendering engine to provide real time visibility into CPU, RAM, and network throughput with sub millisecond latency.
- Implemented a non blocking event polling system and a persistent tracker struct to calculate network deltas and dynamic process prioritization without interrupting the 500ms refresh cycle.
- Developed a memory safe hardware abstraction layer using Rust's ownership model to manage terminal raw mode, alternate screen buffers, and graceful system restoration upon application exit.

Echo [↗](#)

Multi-tenant SaaS Storage Engine & IAM Framework

Github [↗](#)

(Node.js, TypeScript, Drizzle ORM, MinIO/S3, Podman)

- Architected a modular monolith using a Service Repository pattern to provide strict data isolation across a multi tenant hierarchy, managing complex relational memberships between Users and Organizations.
- Engineered an S3 compatible object storage layer utilizing Multer memory-storage to stream binary data directly to a containerized MinIO instance, bypassing local filesystem persistence to ensure horizontal scalability.
- Implemented a secure asset retrieval system leveraging time limited cryptographic presigned URLs and Version 4 UUID key obfuscation to prevent unauthorized access and physical filename exposure.
- Developed a type safe metadata tracking engine with Drizzle ORM and PostgreSQL to manage MIME type validation, file ownership, and organizational partitioning via foreign key constraints.
- Optimized local first development infrastructure on Linux using Podman native containerization and SELinux labeled volumes to maintain environmental parity with production cloud storage environments.

Kaen [↗](#)

Real time Community Platform for Developers

Github [↗](#)

(React, TypeScript, Supabase, TanStack Query)

- Built a high-performance discussion engine featuring real-time post synchronization via PostgreSQL replication, an instant global search with optimized filtering, and a recursive nested comment architecture for complex threading.
- Implemented a robust engagement system including a debounced upvoting/liking mechanism to minimize API overhead, dynamic comment counts, and granular data security through Supabase Row Level Security (RLS).
- Designed a responsive "HUD-style" interface utilizing Framer Motion's hardware-accelerated transitions, featuring adaptive Grid/List view variants and mobile-first metadata optimization.

Archfolio [↗](#)

Operating System Interactive Portfolio

Github [↗](#)

(React, GSAP, Tailwind CSS, JavaScript)

- Built a full-scale virtual desktop environment featuring a window management system, draggable GUI components, and a custom-built file system architecture.
- Implemented a dual-mode navigation engine including a functional terminal emulator with command-line parsing and a graphical file manager for directory traversal.
- Designed a high-performance animation system using GSAP for window logic and state-driven UI transitions, ensuring a seamless user experience across modular components.

Sven

Real time Language Exchange Platform

Github [↗](#)

(React, Node.js, Express.js, React Query, Axios)

- Engineered a real-time communication platform facilitating interactive language exchange through live chat and video call integration, focusing on low-latency data synchronization.
- Developed a robust state management layer using React Query and Axios to handle asynchronous data fetching and real-time UI updates across complex learning workflows.
- Architected a scalable RESTful backend to manage user authentication, immersive conversation sessions, and structured practice modules for a seamless peer-to-peer experience.

Vex [↗](#)

AI Tutor SaaS Platform

Github [↗](#)

(Next.js, TypeScript, Supabase, Stripe)

- Built a full-stack SaaS platform that allows users to create AI tutors and chat with them in real time.
- Implemented secure authentication, protected routes, and subscription-based payments using Stripe.
- Designed a scalable architecture with modular components, server/client separation, and responsive UI using Tailwind CS